SETS

Mathematics Revision Cheatsheet

Secondary School ▪ Basics → Venn-diagram problems

PrepNAce

Mathematics Revision Series

1 What is a set?

A **set** is a well-defined collection of distinct objects called **elements** (members), written in curly brackets: $A = \{1, 2, 3\}$. Order and repeats do not matter: $\{1, 2, 3\} = \{3, 2, 1\} = \{1, 1, 2, 3\}$.

$\in$	is an element of ($2 \in A$)	$\subseteq$	subset (may be equal)
$\notin$	is not an element of	$\subset$	proper subset (not equal)
$n(A)$	number of elements	$\cup$	union: "or"
$\varnothing$, $\{\,\}$	empty set	$\cap$	intersection: "and"
ξ (or $\mathcal{U}$)	universal set	A'	complement of A
$\mathcal{P}(A)$	power set	$A \setminus B$	A without B

2 Ways to describe a set

Listing $A = \{2, 3, 5, 7, 11\}$

Set-builder $A = \{x : x \text{ is a prime}, \ x < 12\}$ ("all x such that ...")

Description "A is the set of primes less than 12"

Example: $B = \{x : x \in \mathbb{N}, \ 2 < x \leq 7\} = \{3, 4, 5, 6, 7\}$, so $n(B) = 5$.

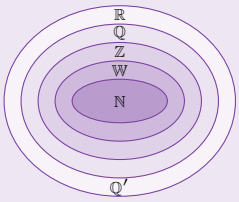
3 Types of sets

Type	Meaning	Example
Finite	elements can be counted	$\{1, 2, 3\}$
Infinite	never ends	$\mathbb{N} = \{1, 2, 3, \dots\}$
Empty (null)	no elements; $n(\varnothing) = 0$	months with 32 days
Singleton	exactly one element	$\{7\}$
Equal	same elements: $A = B$	$\{1, 2\} = \{2, 1\}$
Equivalent	same number of elements	$\{a, b\}$ and $\{1, 2\}$
Disjoint	nothing in common: $A \cap B = \varnothing$	$\{1, 2\}$, $\{3, 4\}$
Overlapping	share some, not all	$\{1, 2, 3\}$, $\{3, 4\}$

✘ $\varnothing \neq \{0\}$ and $\varnothing \neq \{\varnothing\}$: each of these has one element.

4 Number sets

$\mathbb{N}$ Natural: $\{1, 2, 3, \dots\}$
$\mathbb{W}$ Whole: $\{0, 1, 2, 3, \dots\}$
$\mathbb{Z}$ Integers: $\{\dots, -2, -1, 0, 1, 2, \dots\}$
$\mathbb{Q}$ Rational: $\frac{a}{b}$, $b \neq 0$ (e.g. $\frac{1}{2}$, -3)
$\mathbb{Q}'$ Irrational: π, $\sqrt{2}$, $\sqrt{3}$
$\mathbb{R}$ Real numbers $= \mathbb{Q} \cup \mathbb{Q}'$



$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$ (each set sits inside the next).
Irrationals ($\mathbb{Q}'$) have decimals that neither end nor repeat.

5 Subsets & power sets

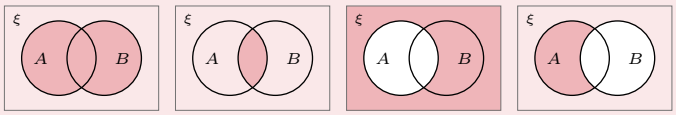
- $A \subseteq B$: every element of A is also in B. Always $\varnothing \subseteq A$ and $A \subseteq A$.
- $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. If $A \subseteq B$ then $A \cup B = B$ and $A \cap B = A$.
- $\in$ joins an *element* to a set; $\subseteq$ joins a *set* to a set: $2 \in \{1, 2, 3\}$ but $\{2\} \subseteq \{1, 2, 3\}$.
- The **power set** $\mathcal{P}(A)$ is the set of all subsets of A.

Set with n elements: 2^n **subsets**, $2^n - 1$ **proper subsets**

n	0	1	2	3	4	5
2^n	1	2	4	8	16	32

Example: $A = \{a, b, c\}$ has $2^3 = 8$ subsets:
$\varnothing$, $\{a\}$, $\{b\}$, $\{c\}$, $\{a, b\}$, $\{a, c\}$, $\{b, c\}$, $\{a, b, c\}$.

6 Set operations & Venn diagrams



$A \cup B$ — A or B (or both)
$A \cap B$ — in both A and B
A' — in ξ, not in A
$A \setminus B$ — in A, not in B

Example: $\xi = \{1, 2, \dots, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$
$A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$ $A \cap B = \{4, 5\}$ $A' = \{6, 7, 8, 9, 10\}$
$A \setminus B = \{1, 2, 3\}$ $B \setminus A = \{6, 7\}$ $n(A') = n(\xi) - n(A) = 5$

7 Laws of sets

Commutative	$A \cup B = B \cup A$ $A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$ (same for $\cap$)
Distributive	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity	$A \cup \varnothing = A$ $A \cap \xi = A$
Domination	$A \cup \xi = \xi$ $A \cap \varnothing = \varnothing$
Complement	$A \cup A' = \xi$ $A \cap A' = \varnothing$ $(A')' = A$
Idempotent	$A \cup A = A$ $A \cap A = A$
De Morgan	$(A \cup B)' = A' \cap B'$ $(A \cap B)' = A' \cup B'$

✔ **De Morgan tip:** split the complement and flip $\cup \leftrightarrow \cap$.

8 Words → symbols

"A *and* B", "both"	$A \cap B$
"A *or* B", "at least one" (inclusive)	$A \cup B$
"not A"	A'
"A only", "A but not B"	$A \setminus B = A \cap B'$
"neither A nor B"	$(A \cup B)' = A' \cap B'$
"exactly one of A, B"	$(A \cup B) \setminus (A \cap B)$

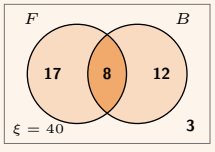
9 Solving Venn-diagram problems

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$
$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$
$$n(A \text{ only}) = n(A) - n(A \cap B) \quad n(\text{neither}) = n(\xi) - n(A \cup B)$$

1. Draw a box for ξ and overlapping circles; label them.
2. Fill the **overlap** $A \cap B$ *first*.
3. Subtract the overlap from each total to get the "only" regions.
4. Outside $= n(\xi) - n(A \cup B)$; check the total is $n(\xi)$.

Example. In a class of 40, 25 play football (F), 20 play basketball (B) and 8 play both. How many play neither?

Only F: $25 - 8 = 17$
Only B: $20 - 8 = 12$
$n(F \cup B) = 25 + 20 - 8 = 37$
Neither $= 40 - 37 = \mathbf{3}$



10 Exam tips & common mistakes

✘ Mixing up $\in$ (element) and $\subseteq$ (subset).
✘ Forgetting $\varnothing$ and the set itself when listing all subsets.
✘ Counting the overlap twice: always subtract $n(A \cap B)$.
✘ Finding a complement without stating what ξ is.
✔ "Or" in maths is inclusive: it includes "both".
✔ List elements in order, with no repeats, and check your totals.